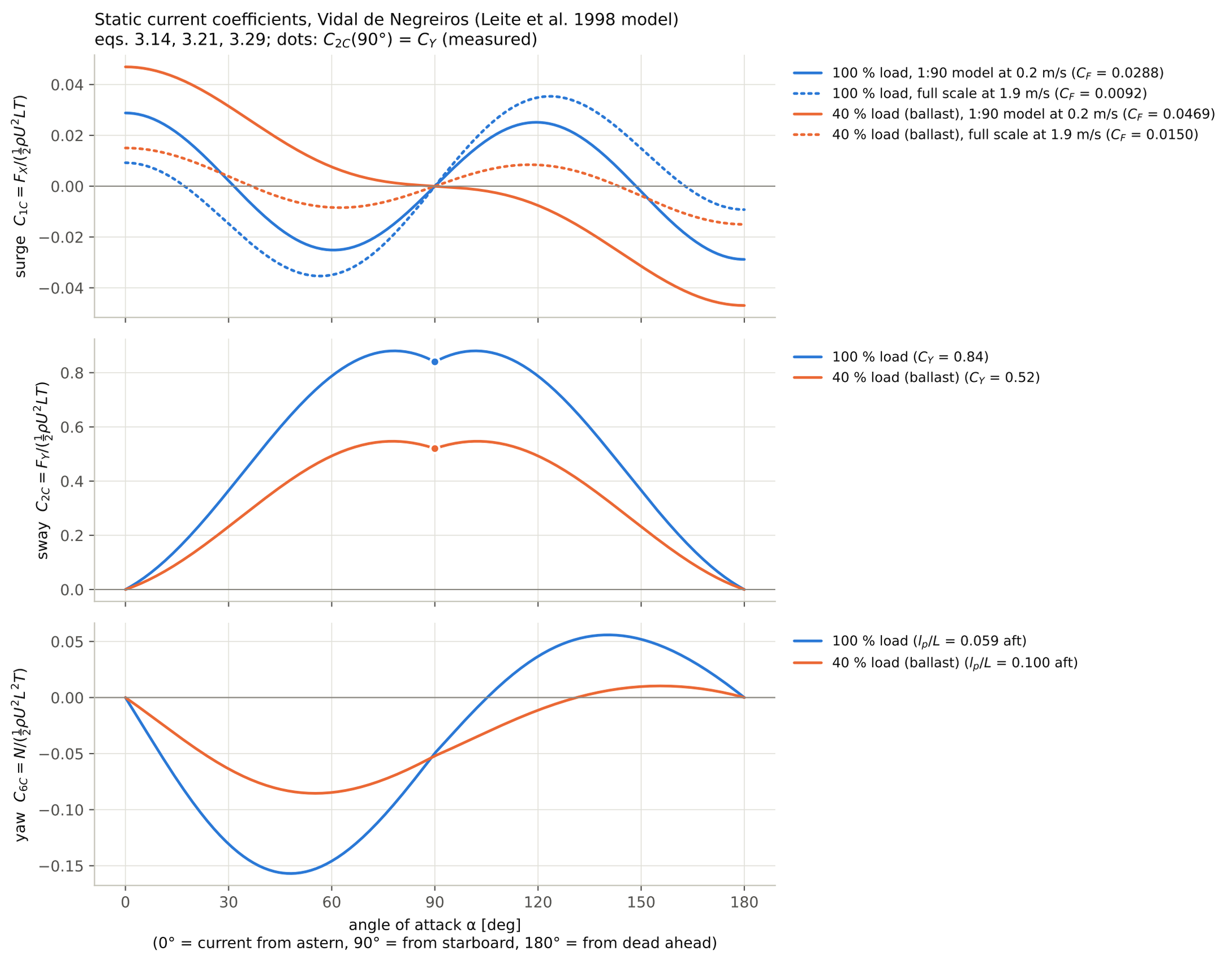
Static current coefficients, Vidal de Negreiros (Leite et al. 1998 model)
eqs. 3.14, 3.21, 3.29; dots: $C_{2C}(90°) = C_Y$ (measured)
surge $C_{1C} = F_X/(\frac{1}{2}\rho U^2 LT)$
sway $C_{2C} = F_Y/(\frac{1}{2}\rho U^2 LT)$
yaw $C_{6C} = N/(\frac{1}{2}\rho U^2 L^2 T)$
0.04
0.02
0.00
−0.02
−0.04
0.8
0.6
0.4
0.2
0.0
0.05
0.00
−0.05
−0.10
−0.15
0
30
60
90
120
150
180
angle of attack α [deg]
(0° = current from astern, 90° = from starboard, 180° = from dead ahead)
100 % load, 1:90 model at 0.2 m/s (C_F = 0.0288)
100 % load, full scale at 1.9 m/s (C_F = 0.0092)
40 % load (ballast), 1:90 model at 0.2 m/s (C_F = 0.0469)
40 % load (ballast), full scale at 1.9 m/s (C_F = 0.0150)
100 % load (C_Y = 0.84)
40 % load (ballast) (C_Y = 0.52)
100 % load (l_p/L = 0.059 aft)
40 % load (ballast) (l_p/L = 0.100 aft)